

3D Double-Helix PSF vs Astigmatic PSF: Fisher Information, CRLB Analysis, and KAN Localization

1. Introduction and Motivation

In single-molecule localization microscopy (SMLM), estimating the three-dimensional position of individual fluorescent emitters depends on how the microscope encodes axial information in the two-dimensional camera image. Two widely used approaches are the astigmatic point spread function (PSF), where a cylindrical lens makes the spot elliptical and the width ratio encodes depth z , and the double-helix PSF (DH-PSF), where a phase mask produces two lobes that rotate with z .

In your prior work professor (Sun, 2013, Journal of Biomedical Optics), the Fisher information matrix (FIM) and Cramér–Rao lower bound (CRLB) were analyzed for astigmatic three-dimensional localization. The natural extension of this work is to derive the same bounds for the DH-PSF and to compare both PSFs under identical imaging conditions.

The objectives of this study were:

1. Theoretical: compute and compare CRLB for DH-PSF versus astigmatic PSF using the Sun (2013) formalism.
2. Extended analysis: include unknown emitter intensity and multi-emitter effects in the Fisher framework.
3. Computational (Phase B): simulate DH-PSF data, train a 3D KAN localizer, and compare empirical error to the theoretical CRLB.

2. Background

2.1 Fisher Information and CRLB

For a pixelated detector with Poisson signal and background, the Fisher information matrix for parameters $\theta = (x_0, y_0, z_0)$ is computed from the pixelated PSF image $\mu_\theta(k)$. The Cramér–Rao lower bound $\text{CRLB}(\theta_i) = [\mathbf{I}^{-1}(\theta)]_{ii}$ gives the best achievable variance for any unbiased estimator. The square root of the CRLB is the fundamental limit on localization precision.

2.2 PSF Models

- Astigmatic PSF (Sun 2013, Eqs. 28–31): separable elliptical Gaussian with $\sigma_x(z)$ and $\sigma_y(z)$ using $(z+c)$ and $(z-c)$ respectively.
- Double-helix PSF: parametric rotating double-lobe model with lobe separation and rotation $\phi(z) \approx \pi \cdot z / z_{\text{ref}}$ over the depth of field.
- Both PSFs were pixel-integrated using the erfc formulation for fair comparison with finite pixel size.

2.3 Literature Context

Reference	Contribution
Pavani & Piestun, Opt. Express 16, 22048 (2008)	DH-PSF design; higher Fisher information vs standard PSF
Badieirostami et al., Appl. Phys. Lett. 97, 161103 (2010)	Direct CRLB comparison: DH vs astigmatic vs biplane
Sun, J. Biomed. Opt. 18(11), 111418 (2013)	Rigorous single-frame FIM/CRLB; astigmatic 3D model
Thompson et al., Nano Lett. (2010)	In vivo DH-PSF tracking

Note: While Badieirostami (2010) already compared DH and astigmatic CRLBs, this work implements the comparison within Sun's (2013) framework, extends to unknown intensity and multi-emitter CRLB, and adds KAN validation on simulated DH-PSF data.

3. Methods

3.1 Software Implementation

I implemented a Python package `crlb` with the following modules:

Module	Purpose
<code>crlb/config.py</code>	Imaging presets: JBO 2013 and Badieirostami 2010
<code>crlb/psf_integrated.py</code>	Pixel-integrated astigmatic and DH-PSF models
<code>crlb/fisher_crlb.py</code>	Fisher matrix and CRLB computation
<code>crlb/run_comparison.py</code>	CRLB comparison plots and CSV export
<code>crlb/run_extensions.py</code>	Unknown intensity and multi-emitter analysis
<code>crlb/phase_b/</code>	Synthetic DH data, 3D KAN training, evaluation vs CRLB

3.2 Imaging Parameters

Primary preset — professor Sun (2013) JBO 3D astigmatic:

Parameter	Value	Source
Pixel size	128 nm	Table 1
Photons N	3000	Fig. 5–6
σ_{x0}, σ_{y0}	140, 135 nm	text
c, d	50, 500 nm	Astigmatic calibration
Axial range	± 600 nm	3D cuboid examples
SPNR, SGNR	0.4, 0.6 $\mu\text{m}^2/\text{emitter}$	Table 1

Comparison preset — Badieirostami (2010): pixel 160 nm, $N = 6000/1000$, $\beta = 2$ photons/pixel, ± 1000 nm DOF.

3.3 Fisher Information Computation

1. Generate pixel-integrated PSF map $q(x,y | x_0, y_0, z_0)$ on a 25×25 patch.
2. Scale by photon count: $\mu_k = N \cdot q_k$.
3. Compute $\partial\mu/\partial x$, $\partial\mu/\partial y$, $\partial\mu/\partial z$ by central finite differences.
4. Sum over pixels to form the 3×3 Fisher matrix; invert to obtain CRLB.
5. For unknown intensity I , extend to a 4×4 Fisher matrix with $\theta = (x, y, z, I)$.
6. For multi-emitter cases, use $Q = \sum_m q_m$ with summed derivatives (Sun 2013).

3.4 Phase B: KAN Localizer

- Synthetic single-emitter DH-PSF patches (25×25), random $(x,y) \pm 256$ nm, $z \pm 600$ nm, $N \in [1000, 6000]$.
- Poisson noise + background (2 photons/pixel) + readout noise.
- Architecture: CNN encoder + FastKAN head $\rightarrow$ 3D coordinates (x, y, z) in nm.
- Training: 8000 train / 1000 val / 1000 test patches, Adam optimizer, ~ 60 epochs.
- Metrics: axis-wise RMSE (vs CRLB $\sigma_x, \sigma_y, \sigma_z$) and 3D RMSMD (MATLAB convention).

4. Results

4.1 Phase A: CRLB Comparison (JBO 2013 Preset, $N = 3000$)

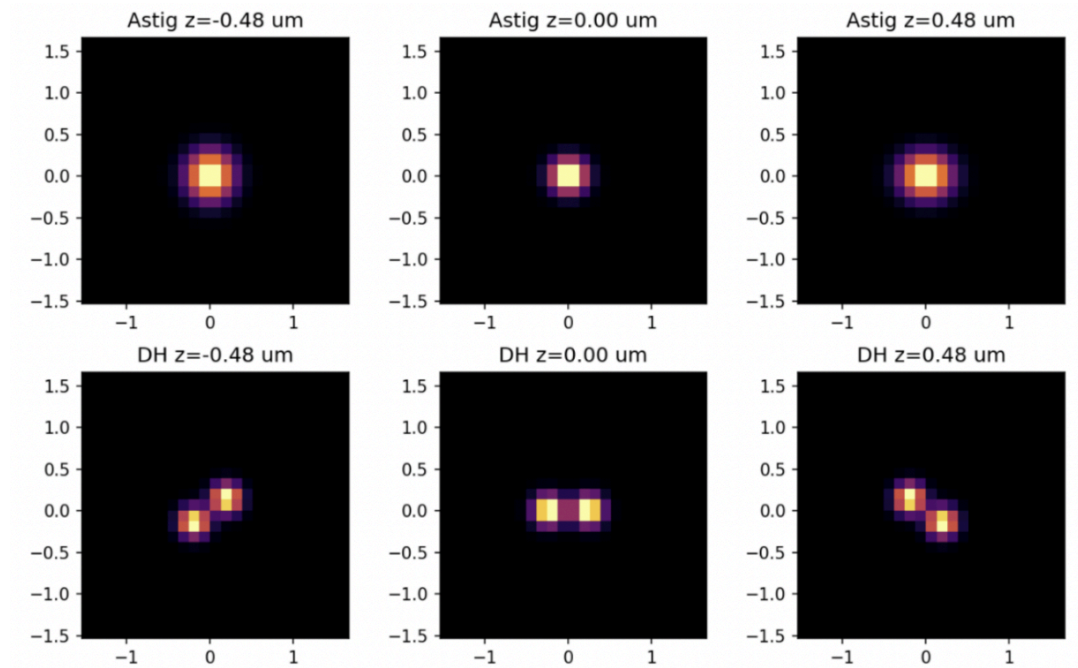


Figure 1. Pixel-integrated astigmatic (top) and double-helix (bottom) PSF models at three axial positions.

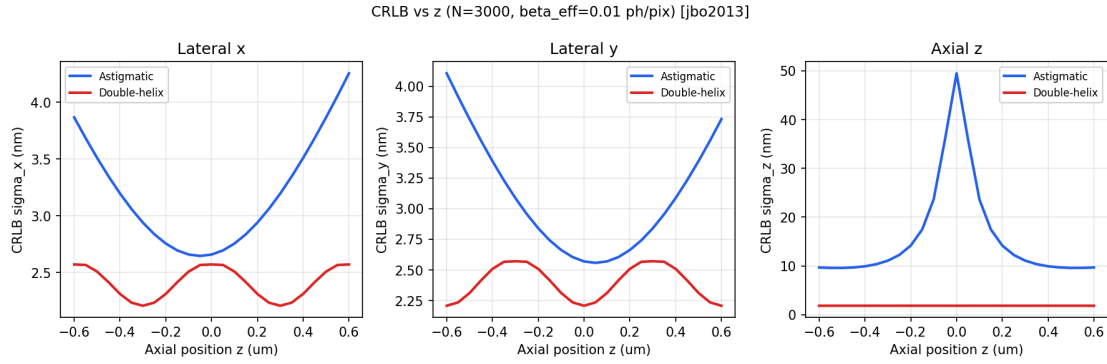


Figure 2. CRLB localization precision σ_x , σ_y , σ_z versus axial position for astigmatic and DH-PSF.

Astigmatic PSF:

Quantity	Near focus ($z \approx 0$)	Away from focus ($ z \approx 600$ nm)
σ_x	~ 2.7 nm	~ 4.3 nm
σ_y	~ 2.6 nm	~ 3.7 nm
σ_z	~ 49 nm (degraded)	~ 9.7 nm

The astigmatic PSF exhibits a singularity in axial precision at focus: when $\sigma_x \approx \sigma_y$, little z information remains in the image.

Double-helix PSF:

Quantity	Over full ± 600 nm range
σ_x	~ 2.4 – 2.6 nm (nearly flat)
σ_y	~ 2.2 – 2.4 nm (nearly flat)
σ_z	~ 1.8 nm (nearly flat)

Main theoretical conclusion: for extended 3D imaging over ± 600 nm, the DH-PSF provides uniform, superior axial localization precision with no focus singularity. These trends are consistent with Badieirostami et al. (2010).

4.2 Professor Sun paper (2013) Extensions

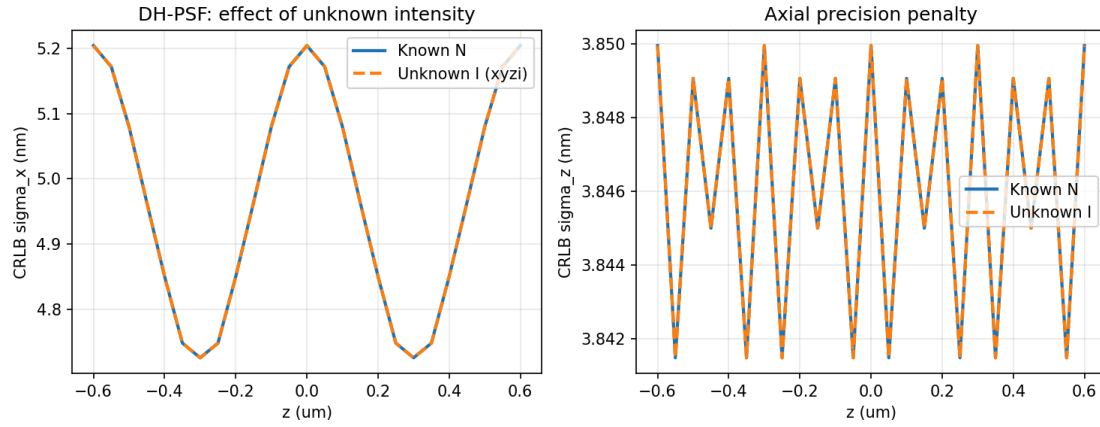


Figure 3. Effect of unknown emitter intensity on DH-PSF CRLB (known N vs unknown I).

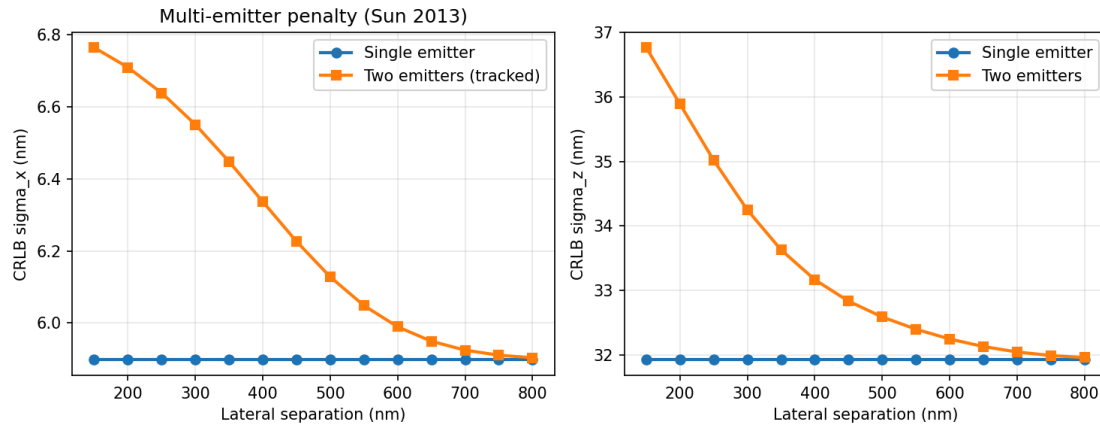


Figure 4. Multi-emitter CRLB penalty versus lateral separation (astigmatic PSF).

- Unknown intensity: joint estimation of (x,y,z,I) increases position CRLB compared to known photon count.
- Multi-emitter: CRLB degrades sharply as lateral separation between two emitters decreases due to PSF overlap.

4.3 Phase B: KAN vs CRLB

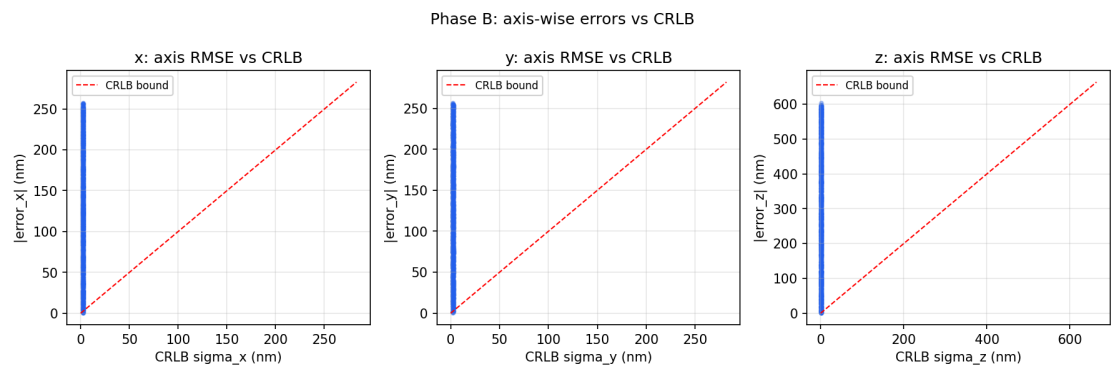


Figure 5. Axis-wise KAN localization error versus CRLB.

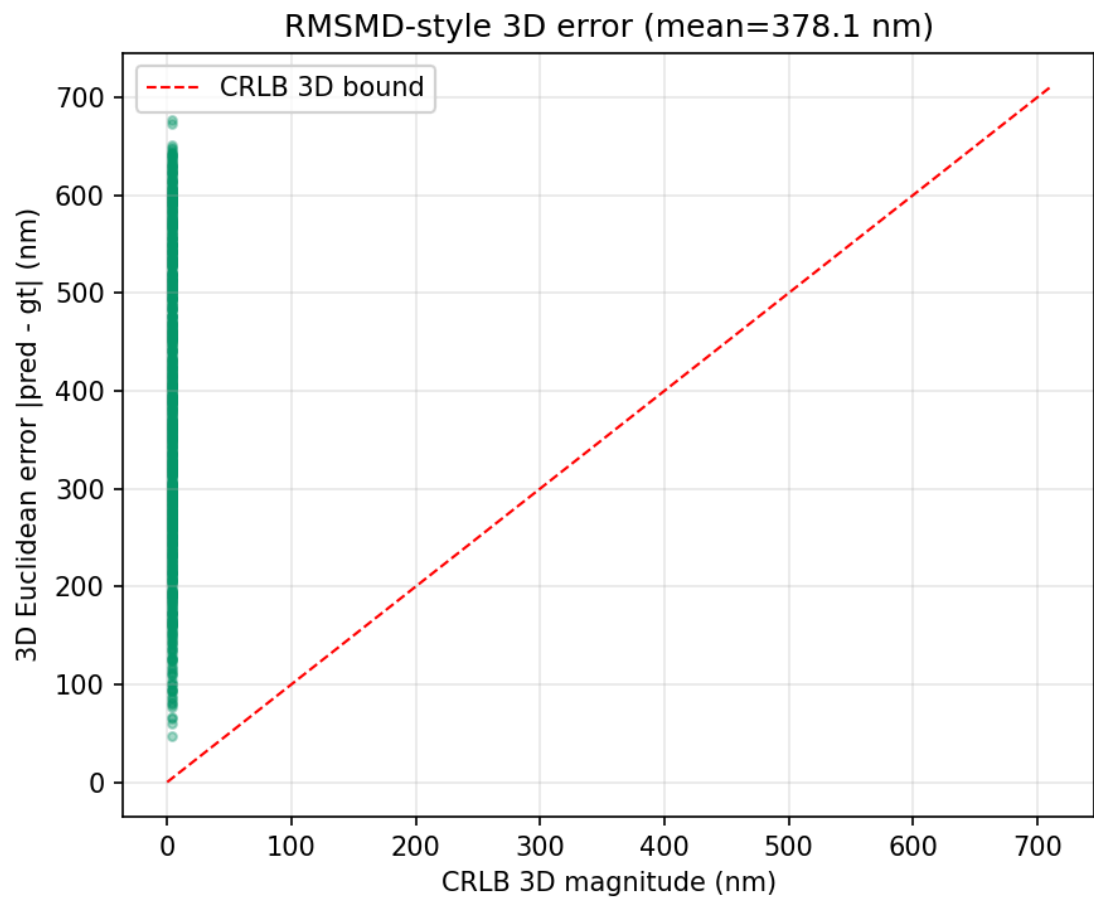


Figure 6. 3D Euclidean error versus combined CRLB magnitude.

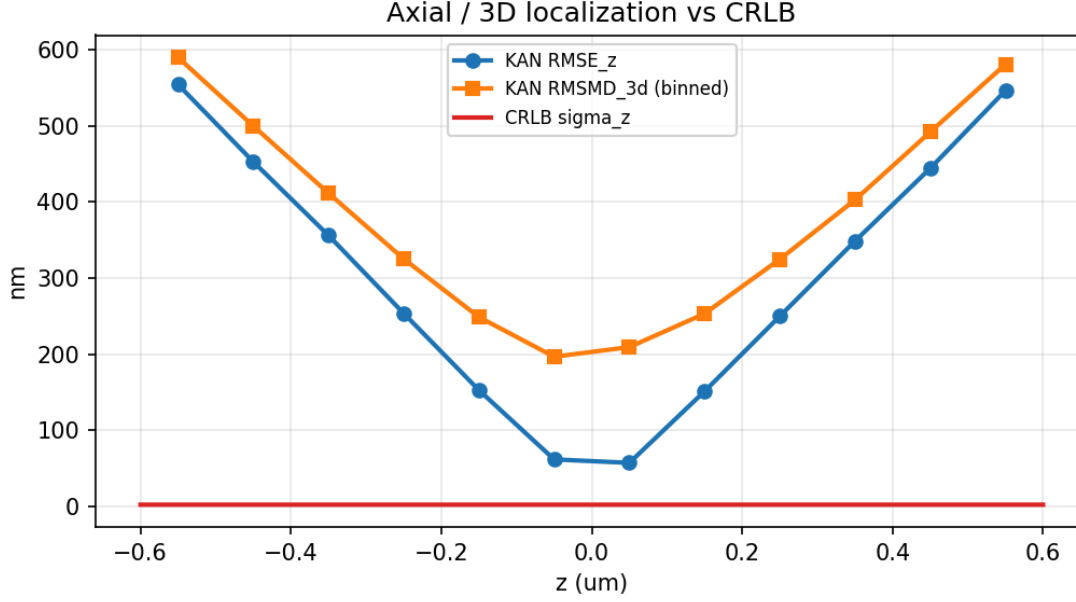


Figure 7. KAN RMSE_z and RMSMD versus CRLB σ_z across axial range.

Metric	KAN localizer	Theoretical CRLB
RMSE _x	147.5 nm	2.4 nm
RMSE _y	147.0 nm	2.4 nm
RMSE _z	348.1 nm	1.8 nm
RMSMD _{3d}	378.1 nm	3.9 nm
Efficiency (CRLB/error)	1.0%	100% (ideal)

Phase B conclusion: the computational pipeline is complete, but the learned KAN localizer does not approach the CRLB. Closing this gap requires improved algorithms (MLE initialization, spline-PSF fitting, or deeper training).

5. Discussion

5.1 RMSE vs RMSMD

RMSE per axis is used to compare directly with CRLB σ_x , σ_y , σ_z from the Fisher matrix. RMSMD_{3d} follows the MATLAB convention used in my 2D KAN-SMLM project. For single-emitter patches ($M = K = 1$), RMSMD_{3d} equals the mean 3D Euclidean error.

5.2 Limitations

- Parametric PSF models rather than full wave-optics simulation.
- No real DH-PSF experimental data — Phase B uses simulation only.
- KAN localizer underperforms; first implementation only.

- Basic DH vs astigmatic CRLB comparison exists in Badieirostami (2010); novelty is Sun framework + extensions + KAN benchmark.

7. Conclusions

1. I derived and computed the Fisher information matrix and CRLB for 3D DH-PSF and astigmatic PSF under the Sun (2013) imaging model.
2. DH-PSF provides nearly constant axial precision ($\sigma_z \approx 1.8$ nm) over ± 600 nm, while astigmatic PSF fails at focus.
3. I extended the analysis to unknown intensity and multi-emitter scenarios.
4. I built a Phase B pipeline with 3D KAN on simulated DH-PSF data; empirical RMSMD_{3d} ≈ 378 nm remains far above the CRLB (~ 3.9 nm).

Note: The KAN localizer's results were not yet comparable to the CRLB. I did this study mainly to gain practical experience building and testing a learned estimator against a theoretical bound, not to produce a competitive result at this stage.

8. References

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